

Hence, additional brackets have to be fixed on the grills. As shown in Fig. 3 and Fig. 4, brackets should be fixed at top and bottom edges of the panels.

Thus, there is a need to introduce building code for fixing the 'guides' for standardisation purpose. This will go a long way in making the solar panel installation simple and safe.

Electrical accessories

The proposed system produces direct current (DC) and so has polarity. Hence, it is necessary to standardise altogether different types of accessories for this system such as pins, sockets, cables, connectors, switches, lamp holders, etc.

The design should be such that a user would be unable to plug in PV accessories into 230V mains supply or vice versa by mistake. Cables also should be colour coded, so that it is easy to identify which is PV cable and which is 230V AC cable. This is very important for user safety as well as for ease of use without any confusion to a layman.

Power levels

User should be able to decide what will be the power rating of the system he wants to install. This decision will depend upon how much money he can invest and how many panels could be mounted in his apartment. The lowest power will be using single panel with 100Wp rating. If more than one panel has to be installed, the power rating will be in multiples of 100Wp.

All the panels have to be connected in parallel to maintain standard voltage of 120V. One Schottky diode should be connected in series with each panel before connecting the panels in parallel. The diode stops one panel dumping current in another panel if the PV voltages are not equal. Do not connect panels in series, as it will produce 240V, which is not safe.

Voltage levels

Open circuit voltage of panel = $21.2 \times 7 = 148.4$, or about 150V

Minimum PV voltage = 100V

Max current for 100W system = $100W/122.5V = 0.816A$. (Cables should be selected for at least 1A rating)

Note that the accessories have to be rated for at least 150V, otherwise insulation breakdown may happen.

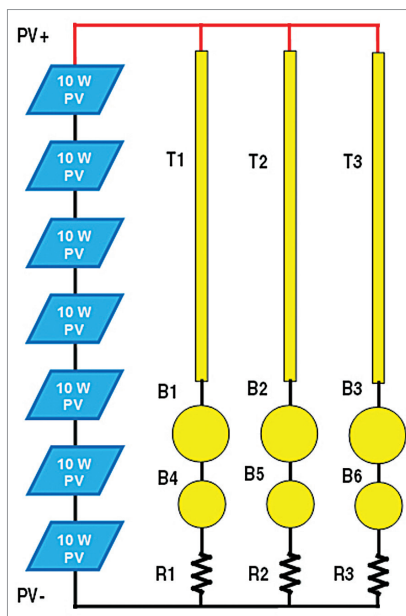


Fig. 5: Simple solar day lamp operating at 120V PV supply

So, check the ratings of the accessories before using them.

Precautions

1. Do not run 230V AC mains and PV cables together in the same conduit. These two wirings have to be kept separate to avoid short circuit between the networks.

2. Do not touch the live 120V wires. Even though 120V does not give a severe shock, it is good practice not to touch these bare wires. If possible, work either in the early mornings or in the evenings/night, when PV voltage is very less/zero.

3. Do not use 230V accessories such as plug tops, sockets, switches, etc. For PV system use altogether different types of switches (toggle switches), jack pins, terminal strips, etc that are not used for 230V AC.

4. Do not short PV + and PV- terminals. Also, do not use any accessories that are not polarised. Otherwise, the supply polarity to the appliance will get reversed.

5. Do not connect mobile charger to V_{pv} and try to charge a mobile phone. It will damage the phone due to fluctuating nature of sunlight.

Solar day lamp

Circuit diagram of a simple solar day lamp (SDL) that works on the 120V PV supply voltage is shown in Fig. 5. Seven



Fig. 6: Photograph of SDL having 3x36 LEDs turned on

10Wp PV panels are connected in series. This is a sample application of the proposed 120V PV system for apartments.

The reason for showing separate panels is that, as of today, single panels that generate 120V are not available in the market. So, for those who want to implement SDL with seven individual panels, they can do so on their roof-top. (Do not install panels made using glass panes on a balcony; it is unsafe.)

Three lamp loads are connected to the PV output. Each lamp load consists of following LED arrays. Each LED is a white LED with 1W power rating:

1. T1, T2, T3→24 LEDs connected in series. This MCPCB is used in tube lights. It is mounted inside a small aluminium 'C' channel. Apply thin layer of heat-sink compound for better heat transfer.

2. B1, B2, B3→7 LEDs connected in series. This MCPCB is a round PCB used in 7W LED bulbs. Mount it on an aluminium channel.

3. B4, B5, B6→5 LEDs connected